Department of Computer Science & Engineering
Microprocessor & Computer Architecture
UE24CS251B

MPCA Mini-project

Title: Smart Fire Detection System	
Team Members:	
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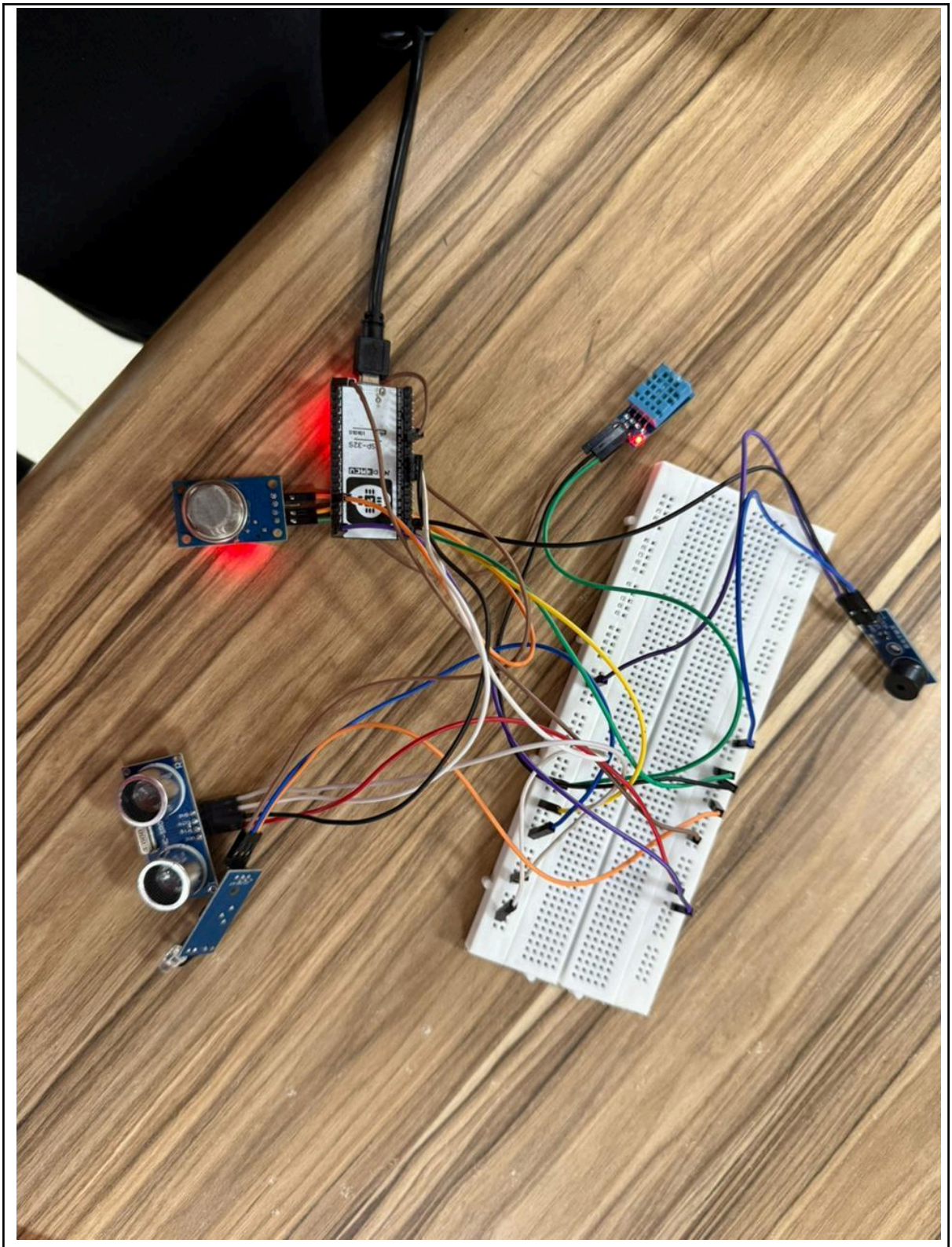
Abstract/Brief description of the project: Smart Fire Detection System is a solution that enhances fire safety through intelligent detection and decision-making. It uses multi-sensor data to accurately identify fire conditions and human presence, and classifies situations into HIGH, LOW, or SAFE priority levels to enable efficient rescue response. The system operates in both online and offline modes, ensuring reliable monitoring and alert generation even in the absence of network connectivity.

Components used:
ESP32 Microcontroller
DHT11 Sensor (Temperature & Humidity)
MQ-135 Gas Sensor
HC-SR04 Ultrasonic Sensor
PIR Motion Sensor
Buzzer
Breadboard
Connecting Wires

Circuit Diagram: